Data Mining & Business Intelligence

Q1. 2-Mark Questions

Q1. What is clustering? (Chapter 4)

Q2. Explain the different distance measures that can be used to compute distance between two clusters. (Chapter 4)

Q3. Difference between classification and clustering. (Chapter 4)

Q4. Types of Clustering Methods? (Chapter 4)

Q5. Explain k-means clustering Algorithm. (Chapter 4)

Q6. Explain k-medoids Clustering? (Chapter 4)

Q7. What is hierarchical clustering? Explain any 2 techniques for finding distance. (Chapter 4)

Q8. Explain Birch algorithm with an example. (Chapter 4)

Q9. Explain DB Scan clustering algorithm with an example. (Chapter 4)

Q10. What is an outlier? Describe methods that are used for outlier detection analysis. (Chapter 4)

Q11. Explain Market Basket Analysis with an example. (Chapter 5)

Q12. Explain frequent item sets, close item sets and large item sets. (Chapter 5)

Q13. Define and explain support and confidence. (Chapter 5)

Q14. What is apriori algorithm. (Chapter 5)

Q15. Explain Multilevel association rule. (Chapter 5)

Q16. Explain Multidimensional Association rule. (Chapter 5)

Q17. Define Business Intelligence system with example. (Chapter 6)

Q18. Explain Business Intelligence Issues. (Chapter 6)

Chapter - 4 ka 5-Mark Questions

Q19. Use K means algorithm to create 3 clusters for given set of values: (2, 3, 6, 8, 9, 12, 15, 18, 22).

Q20. Use K means to cluster the following data sets into 3 clusters:

Protein	20	21	15	22	20	25	26	20	18	20
Fat	9	9	7	17	8	12	14	9	9	9

Q21. Suppose we have six objects with name A, B, C, D, E, F. Apply single linkage clustering and draw a dendrogram for the given data:

Object	X	Y
A	1	1
B	1.5	1.5
C	5	5
D	3	4
E	4	4
F	3	3.5

Chapter - 5 ka 5-Mark Questions

Q22. Consider the transaction database given below. Apply apriori algorithm with minimum support of 50% and confidence of 50%. Find all Frequent itemsets and all the association rules.

Tid	Items
100	1,3,4
200	2,3,5
300	1,2,3,5
400	2,5

Q23. Use the Apriori algorithm to identify the frequent item-sets in the following database. Then extract the strong association rules from these sets.

Minimum support - 30%, Minimum confidence = 75%.
MU - May 16. Dec. 16

TID.	ITEMS
01	A B. D. E. P
02	B, C. E.
04	A B. D, E,
04	A. B. C, E.
05	A. B. C. D. E. F
06	B C. D
07	A,B, D, E

Q24. What are multidimensional association rules? Explain with suitable examples.

Chapter - 6 ka 5-Mark Questions

Q26. List and explain definition of decision support system with a diagram.

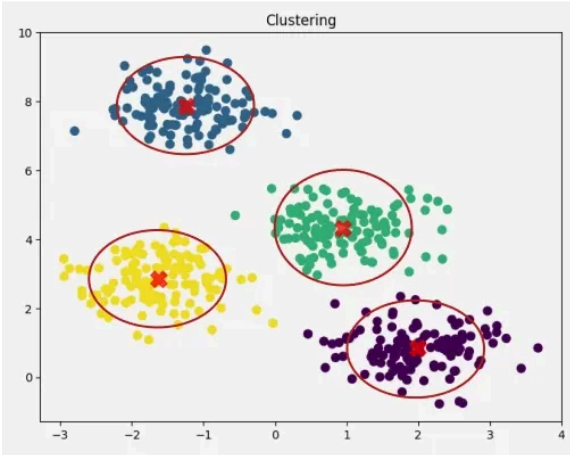
Q27. Explain development of BI system with architecture.

Q1. ANSWERS For 2-Mark Questions

Q1. What is clustering?

- It is a **data mining technique** that groups data objects based on similarity.

- Each cluster contains objects that are **similar to each other**.
- Objects in different clusters are **dissimilar**.
- Commonly used for **pattern recognition, market segmentation, and data analysis**.



Clustering

Q2. Explain the different distance measures that can be used to compute distance between two clusters.

- **Single Linkage (Minimum Distance):** Distance between the **closest pair** of points in two clusters.
- **Complete Linkage (Maximum Distance):** Distance between the **farthest pair** of points in two clusters.
- **Average Linkage:** Distance is the **average of all pairwise distances** between points in the two clusters.
- **Centroid Distance:** Distance between the **centroids (mean points)**

Q3. Difference between classification and clustering.

Aspect	Classification	Clustering
Definition	Assigns data to **predefined classes/labels**	Groups data into clusters **without labels**
Supervision	**Supervised learning** (uses training data)	**Unsupervised learning** (no prior labels)
Goal	Predict class/category of new data	Discover natural groupings/patterns
Example	Classifying emails as spam or not spam	Grouping customers by purchasing behavior

Q4. Types of Clustering Methods?

- **Partitioning Methods:** Divide data into non-overlapping clusters (e.g., k-means, k-medoids).
- **Hierarchical Methods:** Build clusters in a tree-like structure (agglomerative or divisive).
- **Density-Based Methods:** Form clusters based on dense regions of data (e.g., DBSCAN).
- **Grid-Based Methods:** Use a grid structure to cluster data (e.g., STING).

Q5. Explain k-means clustering Algorithm.

- **Step 1:** Choose the number of clusters k.
- **Step 2:** Initialize k centroids randomly.
- **Step 3:** Assign each data point to the **nearest centroid** (forming clusters).
- **Step 4:** Recalculate centroids as the **mean of points** in each cluster.
- **Step 5:** Repeat assignment and update until centroids stabilize (no change).

Q6. Explain k-medoids Clustering?

- Similar to **k-means**, but instead of centroids, it uses **medoids** (actual data points) to represent clusters.
- A **medoid** is the most centrally located point in a cluster, minimizing total dissimilarity.
- Each data point is assigned to the **nearest medoid**.
- More **robust to noise and outliers** compared to k-means.

Q7. What is hierarchical clustering? Explain any 2 techniques for finding distance.

Hierarchical Clustering:

- It is a clustering method that builds a **tree-like structure (dendrogram)** of nested clusters.
- Clusters are formed either by **agglomerative (bottom-up)** merging or **divisive (top-down)** splitting.

Two Techniques for Finding Distance:

- **Single Linkage (Minimum Distance):** Distance between the **closest pair of points** in two clusters.
- **Complete Linkage (Maximum Distance):** Distance between the **farthest pair of points** in two clusters.

Q8. Explain Birch algorithm with an example.

BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) is a clustering algorithm designed for very large datasets. It builds a compact summary of the data using a CF (Clustering Feature) tree, then performs clustering on this summary. It is efficient, incremental, and handles noise well.

Key Points about BIRCH Algorithm

- **Purpose:** Efficient clustering of large datasets with limited memory and time.
- **Approach:** Builds a **CF Tree** (a hierarchical structure) that stores compact summaries of data points.
- **Process:**
  1. Insert data points into the CF tree.
  2. Condense the tree by removing outliers/noise.
  3. Apply a global clustering algorithm (like k-means) on the reduced dataset.

Example:

Suppose we have **10,000 customer transaction records**:

- Instead of clustering all 10,000 directly, BIRCH builds a **CF tree** summarizing them into, say, **100 compact nodes**.
- Then clustering (e.g., k-means) is applied on these 100 nodes instead of the full dataset.
- Result: Faster clustering with similar accuracy, even under memory constraints.

Q9. Explain DB Scan clustering algorithm with an example.

- **Definition:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups points that are **closely packed** together, marking points in low-density regions as **outliers/noise**.
- **Key Idea:** Uses two parameters – **ε (epsilon)** for neighborhood radius and **MinPts** for minimum points required to form a dense region.
- **Process:**
    - Points with enough neighbors → **Core points**.
    - Points within ε of a core → **Reachable points**.
    - Points not reachable → **Noise/outliers**.
- **Example:** In customer data, DBSCAN can detect **dense clusters of frequent buyers** while treating irregular buyers as noise.

Q10. What is an outlier? Describe methods that are used for outlier detection analysis.

Outlier:

- An **outlier** is a data point that significantly **differs from other observations** in a dataset.
- It may indicate **errors, rare events, or unusual patterns**.

Methods for Outlier Detection:

- **Statistical Methods:** Use mean, standard deviation, or z-scores to detect extreme values.
- **Distance-Based Methods:** Identify points far from others using distance measures.
- **Density-Based Methods:** Detect points in low-density regions (e.g., DBSCAN).
- **Clustering Methods:** Points not fitting well into any cluster are treated as outliers.

Q11. Explain Market Basket Analysis with an example.

- It is a **data mining technique** used to find **associations or co-occurrences** among items purchased together.
- Based on **association rules** like *"If item A is bought, item B is likely to be bought."*
- Helps in **cross-selling, promotions, and product placement**.
- **Example:** If customers often buy *bread* and *butter* together, a store may place them nearby or offer combo discounts.

Q12. Explain frequent item sets, close item sets and large item sets.

- **Frequent Item Sets:** Item sets that appear in a dataset with **support ≥ minimum threshold**.
- **Closed Item Sets:** Frequent item sets that have **no superset with the same support count**.
- **Large Item Sets:** Item sets that satisfy the **minimum support condition** (similar to frequent item sets, often used interchangeably).
- **Example:** In transaction data, {bread, butter} is frequent if it appears often; if no larger set has the same frequency, it is closed; and if support ≥ threshold, it is large.

Q13. Define and explain support and confidence.

- **Support:** Measures how frequently an item set appears in the dataset.
  - Example: If {bread, butter} appears in 30% of transactions, its support = 30%.
- **Confidence:** Measures the strength of an association rule, i.e., the probability that item B is bought when item A is bought.
  - Example: Rule {bread → butter} has confidence = (transactions with bread & butter) ÷ (transactions with bread).

Q14. What is apriori algorithm.

- It is an **association rule mining algorithm** used to find **frequent item sets** in transactional data.
- Works on the principle that **all subsets of a frequent item set must also be frequent**.
- Uses **support and confidence** to generate strong association rules.
- Example: If {milk, bread} is frequent, Apriori can derive the rule *milk → bread*.

Q15. Explain Multilevel association rule.

- These are **association rules** that explore relationships at **different levels of abstraction** in a hierarchy.
- Example: Rule at higher level → *"If milk is bought, bread is bought."*
- At lower level → *"If Amul milk is bought, Britannia bread is bought."*
- Useful for **retail analysis**, where items are organized in categories/subcategories.

Q16. Explain Multidimensional Association rule.

- These rules describe **associations across multiple attributes or dimensions**, not just items.
- They can involve **different fields** like time, location, or customer type along with items.
- **Example:** *"Young customers in Mumbai buying laptops also buy headphones."* (Here, age, location, and product are different dimensions).
- Useful in **complex data analysis** where multiple factors influence buying behavior.

Q17. Define Business Intelligence system with example.

- A **Business Intelligence (BI) system** is a technology-driven process for **collecting, analyzing, and presenting business data** to support decision-making.
- It helps organizations turn **raw data into meaningful insights**.
- Provides tools like **dashboards, reports, and data visualization** for managers.
- **Example:** A retail company using BI to analyze sales data and identify the **best-selling products** for inventory planning.

Q18. Explain Business Intelligence Issues

- **Data Quality Issues:** Incomplete, inconsistent, or inaccurate data reduces reliability of BI insights.
- **Integration Issues:** Difficulty in combining data from multiple sources (databases, applications).
- **Cost & Complexity:** High implementation and maintenance costs for BI systems.
- **User Adoption:** Employees may resist using BI tools due to lack of training or complexity.

Chapter - 4 ka 5-Mark Questions

Q19. Use K means algorithm to create 3 clusters for given set of values: (2, 3, 6, 8, 9, 12, 15, 18, 22).

Step 1: Choose Initial Centroids

Let's pick **2, 9, 18** as the initial centroids.

Step 2: Assign Points to Nearest Centroid

- Cluster 1 (centroid = 2): {2, 3, 6}
- Cluster 2 (centroid = 9): {8, 9, 12}
- Cluster 3 (centroid = 18): {15, 18, 22}

Step 3: Recalculate Centroids

- Cluster 1 mean = (2+3+6)/3 = **3.67 ≈ 4**
- Cluster 2 mean = (8+9+12)/3 = **9.67 ≈ 10**
- Cluster 3 mean = (15+18+22)/3 = **18.33 ≈ 18**

Step 4: Final Clusters (after convergence)

- **Cluster 1:** {2, 3, 6} → centroid ≈ 4
- **Cluster 2:** {8, 9, 12} → centroid ≈ 10
- **Cluster 3:** {15, 18, 22} → centroid ≈ 18

Final Answer:
Using K-Means with k=3, the dataset is divided into clusters:

- Cluster 1: {2, 3, 6}
- Cluster 2: {8, 9, 12}

- Cluster 3: {15, 18, 22}

Q20. Use K means to cluster the following data sets into 3 clusters:

K-Means Clustering for Given Dataset (Protein & Fat values, 3 clusters)

We have **10 categories** with two attributes: **Protein** and **Fat**.
Dataset:

- Category 1: (20, 9)
- Category 2: (21, 9)
- Category 3: (15, 7)
- Category 4: (22, 17)
- Category 5: (20, 8)
- Category 6: (25, 12)
- Category 7: (26, 14)
- Category 8: (20, 9)
- Category 9: (18, 9)
- Category 10: (20, 9)

Step 1: Choose Initial Centroids

Let's pick three starting points:

- C1 = (20, 9)
- C2 = (22, 17)
- C3 = (26, 14)

Step 2: Assign Points to Nearest Centroid

- **Cluster 1 (C1):** (20,9), (21,9), (20,8), (20,9), (18,9), (20,9), (15,7)
- **Cluster 2 (C2):** (22,17)
- **Cluster 3 (C3):** (25,12), (26,14)

Step 3: Recalculate Centroids

- Cluster 1 mean = (20+21+20+20+18+20+15)/7 , (9+9+8+9+9+9+7)/7
  = (134/7 , 60/7) ≈ (19.14 , 8.57)
- Cluster 2 mean = (22,17)
- Cluster 3 mean = ((25+26)/2 , (12+14)/2) = (25.5 , 13)

Step 4: Final Clusters (after convergence)

- **Cluster 1:** {Category 1, 2, 3, 5, 8, 9, 10} → centroid ≈ (19.14, 8.57)
- **Cluster 2:** {Category 4} → centroid = (22,17)
- **Cluster 3:** {Category 6, 7} → centroid ≈ (25.5, 13)

Final Answer:
Using K-Means with k=3, the dataset is divided into clusters:

- **Cluster 1:** Categories {1, 2, 3, 5, 8, 9, 10}
- **Cluster 2:** Category {4}
- **Cluster 3:** Categories {6, 7}

Method-2 to solve the above question:

Q.20]

→ Step 1 : Choosing Initial centroids.

category→1 category 6

$C_1 = (20, 9)$, $C_2 = (25, 12)$

$C_3 = (22, 17)$ → category 4

Step 2 : Calculating for other categories to know their distance from centroid.

Using the Euclidean distance formula:

$$d = \sqrt{(\underbrace{x_2}_{\text{observed value}} - \underbrace{x_1}_{\text{centroid value}})^2 + (\underbrace{y_2}_{\text{obsv value}} - \underbrace{y_1}_{\text{cent val}})^2}$$

Evaluating the distance from centroid to make clusters for next categories

category 2
$C_4 = \sqrt{(21-20)^2 + (9-9)^2}$

using C1 centroid $= 1$

Now, using the centroid 2 values

$$\text{cat } 2 = \sqrt{(21-25)^2 + (9-12)^2}$$
$$= 13$$

using centroid 3 values

$$\text{Cat } 3 = \sqrt{(21-22)^2 + (9-17)^2} = 65$$

So, category 2 belongs to cluster 1 because it is close to it by distance "1" that we have evaluated before.

Cluster 1 : (category 1, category 2)

Step 3 : Updating the centroid for cluster - 1 by evaluating mean.

$$\frac{20+21}{2} = 20.5$$

$$\frac{9+9}{2} = 9$$

Now, c1 changes to

$C_1 = (20.5, 9)$ $C_2 = (25, 12)$

$C_3 = (22, 17)$

Follow the same steps to know distance of other categories w.r.t centroid to form the clusters.

Q21. Suppose we have six objects with name A, B, C, D, E, F. Apply single linkage clustering and draw a dendrogram for the given data:

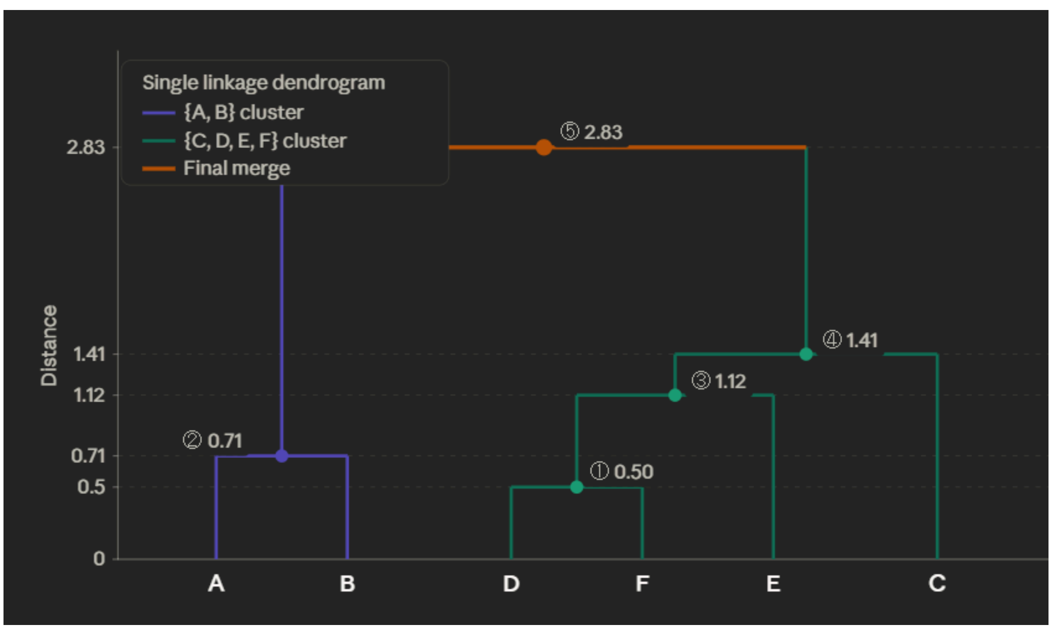
Step 1 – Euclidean distance matrix

Using d= $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

	A	B	C	D	E	F
A	0	0.71	5.66	3.61	4.24	3.54
B	0.71	0	4.95	2.92	3.54	2.83
C	5.66	4.95	0	2.24	1.41	2.50
D	3.61	2.92	2.24	0	1.41	0.50
E	4.24	3.54	1.41	1.41	0	1.12
F	3.54	2.83	2.50	0.50	1.12	0

Step 2 – Single linkage merges (always merge the pair with the *minimum* distance; inter-cluster distance = min of any two members)

Step	Merge	Distance
1	**D + F**	0.50
2	**A + B**	0.71
3	**E + {D,F}**	min(E-D, E-F) = min(1.41, 1.12) = **1.12**
4	**C + {D,E,F}**	min(C-D, C-E, C-F) = min(2.24, 1.41, 2.50) = **1.41**
5	**{A,B} + {C,D,E,F}**	min of all cross-distances = min(B-D=2.92, B-F=2.83, ...) = **2.83**



Reading the dendrogram — step by step:

Step 1 — D and F merge at distance **0.50** (the closest pair in the entire dataset). They're at (3, 4) and (3, 3.5), just half a unit apart.

Step 2 — A and B merge at distance **0.71**. Both sit near (1, 1) and (1.5, 1.5), forming an early tight cluster.

Step 3 — E joins {D, F} at distance **1.12**. The minimum distance from E to any member of {D, F} is E–F = 1.12 (single linkage picks the *nearest* neighbor rule).

Step 4 — C joins {D, E, F} at distance **1.41**. The closest member of that cluster to C is E, at distance 1.41.

Step 5 — The two remaining clusters {A, B} and {C, D, E, F} merge at distance **2.83** (B is the closest point in {A,B} to any point in the other cluster — B–F = 2.83).

Chapter - 5 ka 5-Mark Questions

Q22. Consider the transaction database given below. Apply apriori algorithm with minimum support of 50% and confidence of 50%. Find all Frequent itemsets and all the association rules.

Setup

Transactions (4 total, so min support = 50% → min count = 2):

TID	Items
100	1, 3, 4
200	2, 3, 5
300	1, 2, 3, 5
400	2, 5

Pass 1 — Candidate 1-itemsets (C_1) → Frequent 1-itemsets (L_1)

Itemset	Count	Frequent?
{1}	2	✓
{2}	3	✓
{3}	3	✓
{4}	1	✗ (pruned)
{5}	3	✓

L_1 = { {1}, {2}, {3}, {5} } — item 4 pruned.

Pass 2 — Candidate 2-itemsets (C_2) → Frequent 2-itemsets (L_2)

Generate all pairs from L_1:

Itemset	Count	Frequent?
{1,2}	1	✗
{1,3}	2	✓
{1,5}	1	✗
{2,3}	2	✓
{2,5}	3	✓
{3,5}	2	✓

L_2 = { {1,3}, {2,3}, {2,5}, {3,5} }

Pass 3 — Candidate 3-itemsets (C_3) → Frequent 3-itemsets (L_3)

Join L_2 with itself (pairs that share a (k-1) prefix):

Candidates generated: {1,2,3}, {1,3,5}, {2,3,5}

Pruning by Apriori property — every subset of size 2 must be in L_2:

- {1,2,3}: needs {1,2} → not in L_2 → ✗ pruned
- {1,3,5}: needs {1,5} → not in L_2 → ✗ pruned
- {2,3,5}: needs {2,3}✓, {2,5}✓, {3,5}✓ → survives

Count {2,3,5}: appears in TID 200 and 300 → count = 2 ✓

L_3 = { {2,3,5} }

No 4-itemsets possible. Algorithm stops.

All Frequent Itemsets

L_1: {1}, {2}, {3}, {5}
L_2: {1,3}, {2,3}, {2,5}, {3,5}
L_3: {2,3,5}

Association Rules (min confidence = 50%)

For each frequent itemset, generate all non-trivial rules A → B where A∪B = itemset and confidence = sup(A∪B)/sup(A) ≥ 50%:

From {1,3} — sup = 2:

Rule	Confidence	Pass?
{1}→{3}	2/2 = 100%	✓
{3}→{1}	2/3 = 67%	✓

From {2,3} — sup = 2:

Rule	Confidence	Pass?
{2}→{3}	2/3 = 67%	✓
{3}→{2}	2/3 = 67%	✓

From {2,5} — sup = 3:

Rule	Confidence	Pass?
{2}→{5}	3/3 = 100%	✓
{5}→{2}	3/3 = 100%	✓

From {3,5} — sup = 2:

Rule	Confidence	Pass?
{3}→{5}	2/3 = 67%	✓
{5}→{3}	2/3 = 67%	✓

From {2,3,5} — sup = 2:

Rule	Confidence	Pass?
{2,3}→{5}	2/2 = 100%	✓
{2,5}→{3}	2/3 = 67%	✓
{3,5}→{2}	2/2 = 100%	✓
{2}→{3,5}	2/3 = 67%	✓
{3}→{2,5}	2/3 = 67%	✓
{5}→{2,3}	2/3 = 67%	✓

Frequent itemsets found (9 total):

- L_1: {1}, {2}, {3}, {5} — item {4} is pruned (support 1/4 = 25%)
- L_2: {1,3}, {2,3}, {2,5}, {3,5}
- L_3: {2,3,5}

Q23. Use the Apriori algorithm to identify the frequent item-sets in the following database. Then extract the strong association rules from these sets.

Transactions (7 total, so min support = 30% → min count = ceil(0.30 × 7) = 2.1 → **count ≥ 3** — actually 30% of 7 = 2.1, so we round up to **≥ 3**):

TID	Items
01	A, B, D, E, F
02	B, C, E
03	A, B, D, E
04	A, B, C, E
05	A, B, C, D, E, F
06	B, C, D
07	A, B, D, E

Pass 1 — C_1 → L_1

Itemset	Count	Frequent?
{A}	5	✓
{B}	7	✓
{C}	4	✓
{D}	5	✓
{E}	6	✓
{F}	2	✗ pruned

L_1 = { {A}, {B}, {C}, {D}, {E} } — {F} pruned (count 2 < 3)

Pass 2 — C_2 → L_2

All pairs from L_1:

Itemset	TIDs containing it	Count	Frequent?
{A,B}	01,03,04,05,07	5	✓
{A,C}	04,05	2	✗
{A,D}	01,03,05,07	4	✓
{A,E}	01,03,04,05,07	5	✓
{B,C}	02,04,05,06	4	✓
{B,D}	01,03,05,06,07	5	✓
{B,E}	01,02,03,04,05,07	6	✓
{C,D}	05,06	2	✗
{C,E}	02,04,05	3	✓
{D,E}	01,03,05,07	4	✓

L_2 = { {A,B}, {A,D}, {A,E}, {B,C}, {B,D}, {B,E}, {C,E}, {D,E} }

Pass 3 — C_3 → L_3

Generate by joining pairs from L_2 that share a (k-1)=2-item prefix after sorting:

Candidates generated:

Candidate	Subsets to check (all ∈ L_2?)	Count	Frequent?
{A,B,D}	{A,B}✓ {A,D}✓ {B,D}✓	01,03,05,07 → **4**	✓
{A,B,E}	{A,B}✓ {A,E}✓ {B,E}✓	01,03,04,05,07 → **5**	✓
{A,D,E}	{A,D}✓ {A,E}✓ {D,E}✓	01,03,05,07 → **4**	✓
{B,C,E}	{B,C}✓ {B,E}✓ {C,E}✓	02,04,05 → **3**	✓
{B,D,E}	{B,D}✓ {B,E}✓ {D,E}✓	01,03,05,07 → **4**	✓
{A,B,C}	{A,B}✓ {A,C}✗ → pruned	—	✗
{A,C,E}	{A,C}✗ → pruned	—	✗
{B,C,D}	{C,D}✗ → pruned	—	✗

L_3 = { {A,B,D}, {A,B,E}, {A,D,E}, {B,C,E}, {B,D,E} }

Pass 4 — C_4 → L_4

Join L_3 pairs sharing a 3-item prefix:

Candidate	Subsets to check	Count	Frequent?
{A,B,D,E}	{A,B,D}✓ {A,B,E}✓ {A,D,E}✓ {B,D,E}✓	01,03,05,07 → **4**	✓
{A,B,C,E}	needs {A,B,C} → not in L_3 → pruned	—	✗
{B,C,D,E}	needs {B,C,D} → not in L_3 → pruned	—	✗

L_4 = { {A,B,D,E} }

No 5-itemsets possible. **Algorithm terminates.**

All Frequent Itemsets

Level	Frequent Itemsets
L_1	{A}, {B}, {C}, {D}, {E}

Level	Frequent Itemsets
L_2	{A,B}, {A,D}, {A,E}, {B,C}, {B,D}, {B,E}, {C,E}, {D,E}
L_3	{A,B,D}, {A,B,E}, {A,D,E}, {B,C,E}, {B,D,E}
L_4	{A,B,D,E}

Strong Association Rules (confidence ≥ 75%)

Confidence = sup(X∪Y) / sup(X). Only rules with confidence ≥ 75% are listed.

From {A,B,D,E} — sup = 4:

Rule	Confidence	Strong?
{A,B,D}→{E}	4/4 = **100%**	✓
{A,B,E}→{D}	4/5 = 80%	✓
{A,D,E}→{B}	4/4 = **100%**	✓
{B,D,E}→{A}	4/4 = **100%**	✓
{A,B}→{D,E}	4/5 = 80%	✓
{A,D}→{B,E}	4/4 = **100%**	✓
{A,E}→{B,D}	4/5 = 80%	✓
{B,D}→{A,E}	4/5 = 80%	✓
{B,E}→{A,D}	4/6 = 67%	✗
{D,E}→{A,B}	4/4 = **100%**	✓
{A}→{B,D,E}	4/5 = 80%	✓
{B}→{A,D,E}	4/7 = 57%	✗
{D}→{A,B,E}	4/5 = 80%	✓
{E}→{A,B,D}	4/6 = 67%	✗

From {A,B,D} — sup = 4 (rules not already covered above):

Rule	Confidence	Strong?
{A,B}→{D}	4/5 = 80%	✓
{A,D}→{B}	4/4 = **100%**	✓
{B,D}→{A}	4/5 = 80%	✓
{A}→{B,D}	4/5 = 80%	✓
{D}→{A,B}	4/5 = 80%	✓

From {A,B,E} — sup = 5:

Rule	Confidence	Strong?
{A,B}→{E}	5/5 = **100%**	✓
{A,E}→{B}	5/5 = **100%**	✓
{B,E}→{A}	5/6 = 83%	✓
{A}→{B,E}	5/5 = **100%**	✓
{B}→{A,E}	5/7 = 71%	✗
{E}→{A,B}	5/6 = 83%	✓

From {A,D,E} — sup = 4:

Rule	Confidence	Strong?
{A,D}→{E}	4/4 = **100%**	✓
{A,E}→{D}	4/5 = 80%	✓
{D,E}→{A}	4/4 = **100%**	✓
{A}→{D,E}	4/5 = 80%	✓
{D}→{A,E}	4/5 = 80%	✓
{E}→{A,D}	4/6 = 67%	✗

From {B,C,E} — sup = 3:

Rule	Confidence	Strong?
{B,C}→{E}	3/4 = 75%	✓
{B,E}→{C}	3/6 = 50%	✗
{C,E}→{B}	3/3 = **100%**	✓
{B}→{C,E}	3/7 = 43%	✗
{C}→{B,E}	3/4 = 75%	✓
{E}→{B,C}	3/6 = 50%	✗

From {B,D,E} — sup = 4:

Rule	Confidence	Strong?
{B,D}→{E}	4/5 = 80%	✓
{B,E}→{D}	4/6 = 67%	✗
{D,E}→{B}	4/4 = **100%**	✓
{B}→{D,E}	4/7 = 57%	✗
{D}→{B,E}	4/5 = 80%	✓
{E}→{B,D}	4/6 = 67%	✗

Frequent itemsets (18 total):
L_1 has 5 items, L_2 has 8 pairs, L_3 has 5 triples, and L_4 yields one 4-itemset {A,B,D,E}. Item {F} is pruned in round 1 since it appears in only 2 of 7 transactions (28.6% < 30%).

25 strong association rules were extracted.

Q24. What are multidimensional association rules? Explain with suitable examples.

- **Definition:**
  Multidimensional association rules are rules that describe relationships among items across **multiple attributes (dimensions)**, not just products. These dimensions can include **time, location, customer demographics, or transaction details** along with items.
- **Key Features:**
  1. Extend traditional association rules beyond a single dimension.
  2. Allow analysis of complex datasets with multiple attributes.

3. Useful for discovering patterns that involve **who, what, when, and where**.
4. Provide deeper insights for business decision-making.

- **Types:**
    - **Intra-dimensional rules:** Associations within the same dimension (e.g., product → product).
    - **Inter-dimensional rules:** Associations across different dimensions (e.g., age → product).
- **Example 1:** *"Young customers in Mumbai buying laptops also buy headphones."*
    - Dimensions: Age (young), Location (Mumbai), Product (laptops, headphones).
- **Example 2:** *"Customers who shop on weekends and buy milk also buy bread."*
    - Dimensions: Time (weekend), Product (milk, bread).
- **Applications:**
    - Retail: Combining customer demographics with purchase behavior.
    - Marketing: Targeted promotions based on age, location, and buying habits.
    - Healthcare: Linking patient age, symptoms, and treatments.

Chapter - 6 ka 5-Mark Questions

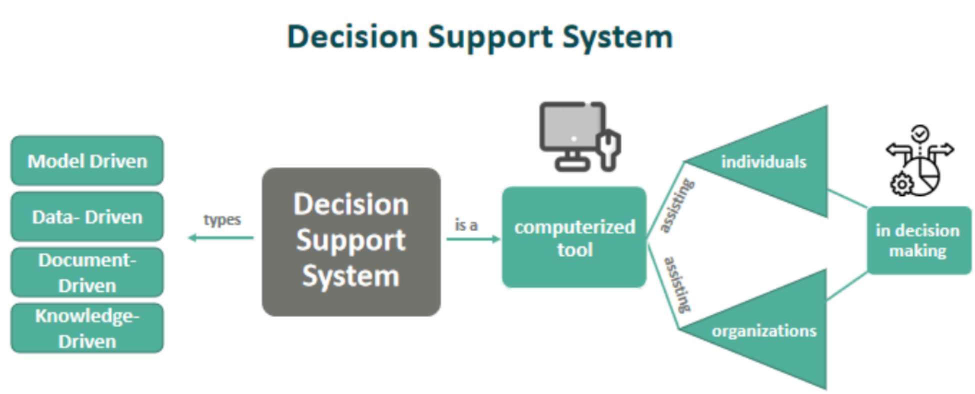
Q26. List and explain definition of decision support system with a diagram.

Decision Support System (DSS) (5 marks)

Definition:
A Decision Support System (DSS) is a **computer-based information system** that helps managers and business professionals make **informed, data-driven decisions**. It combines **data, models, and analytical tools** to support semi-structured or unstructured decision-making.

Key Features:

1. **Data Management:** Collects and organizes data from multiple sources.
2. **Model Management:** Provides analytical models (e.g., forecasting, optimization).
3. **User Interface:** Easy interaction for decision-makers.
4. **Support for Decisions:** Assists in solving problems that are not routine.



Example:

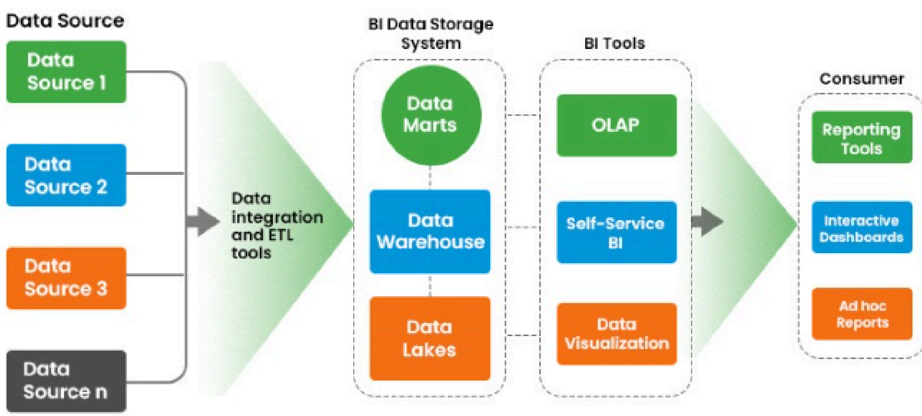
A retail company uses DSS to analyze **sales trends, customer preferences, and inventory levels**. Based on this, managers decide which products to stock more during festive seasons.

Q27. Explain development of BI system with architecture.

Definition:
A Business Intelligence (BI) system is developed to **collect, integrate, analyze, and present business data** for better decision-making. It transforms raw data into actionable insights through systematic stages.

Development Stages:

1. **Data Collection:** Gather data from multiple sources (databases, ERP, CRM, external files).
2. **Data Integration:** Clean, transform, and consolidate data into a central repository.
3. **Data Storage:** Store processed data in a **Data Warehouse** for easy access.
4. **Data Analysis:** Apply OLAP, data mining, and analytical models to extract patterns.
5. **Presentation:** Use dashboards, reports, and visualization tools for decision support.
6. **Decision Making:** Managers use insights for strategic, tactical, and operational decisions.



Example:

A retail chain develops a BI system to integrate **sales, inventory, and customer data**. Managers analyze seasonal trends and customer preferences, then decide which products to stock more during festive seasons.